

Dr. Dinko Milaković

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SCIENTIFIC INTERESTS

Fundamental physics & cosmology	<i>Fundamental constants, redshift drift, gas at high redshift.</i>
Astronomical observations	<i>High resolution optical / near infra-red spectroscopy (HARPS & ESPRESSO).</i>
Improving spectroscopic performance	<i>Laser frequency combs, data reduction, instrument calibration.</i>
Artificial Intelligence	<i>Machine learning, computer vision, deep learning.</i>
Statistical methods	<i>Information criteria, non-linear least squares.</i>

EMPLOYMENT HISTORY

Postdoctoral researcher	Astronomical Observatory of Trieste	<i>since Oct 2024</i>
Postdoctoral researcher	Institute for Fundamental Physics of the Universe (IFPU)	<i>Oct 2021 – Sep 2024</i>
Postdoctoral researcher	Astronomical Observatory of Trieste	<i>Dec 2020 – Sep 2021</i>
Associate	European Southern Observatory (ESO)	<i>Apr 2020 – Sep 2020</i>
Doctoral researcher	European Southern Observatory	<i>Sep 2016 – Mar 2020</i>
Physics tutor	Private teaching	<i>Apr 2015 – Aug 2016</i>

EDUCATION

Ludwig-Maximillian University	PhD in Astronomy Magna cum laude	<i>Sep 2016 – Nov 2020</i>
University of Zagreb	MSc in Physics	<i>Sep 2009 – Mar 2015</i>
University of New South Wales	Exchange student (MSc thesis work)	<i>Jul 2014 – Jan 2015</i>

RESEARCH HIGHLIGHTS

HARPS as a testbed for new technology (3 peer-reviewed publications)

- Developed new methods for wavelength calibration based on astronomical laser frequency comb (LFC) technology and successfully applied them to archival LFC observations taken using the ESO 3.6m/HARPS spectrograph.
- Independently built a data analysis pipeline and investigated the practical limits of using LFCs.
- By comparing two LFCs operated simultaneously, I established that long-term radial velocity stability with LFCs is 45 cm s^{-1} at best, significantly above the photon noise limit (3 cm s^{-1}). This result motivated an experiment that compared the LFC calibration to a calibration from an iodine cell on ESPRESSO (see below).
- By reconstructing the line-spread function (LSF) of HARPS, I demonstrated that using accurate LSF models results in spectroscopic accuracy close to the photon noise limit.
- I am developing (in collaboration with B. Bassett) deep learning methods to reconstruct the point-spread function (PSF) from LFC spectra. These methods will be used to develop a new data reduction pipeline for HARPS based on “spectral perfectionism”, a testbed for a future ELT/ANDES pipeline based on the same principles.

Artificial Intelligence and Fundamental Physics (8 peer-reviewed publications)

- Co-developed AI-VPFIT, a new data analysis tool combining non-linear least squares with an information criterion, Monte Carlo methods, genetic and artificial intelligence algorithms to automatically produce unbiased, reproducible, and robust models of complex spectroscopic datasets. AI-VPFIT presents a major leap in the field, speeding up the modelling process by a factor of >100 when used on a supercomputer, freeing up human resources for other tasks.
- Carried out design studies of AI-VPFIT during its early development. I tested AI-VPFIT using turbulent broadening and made 2 important (closely related) discoveries: (i) turbulent broadening resulted in too many absorption components and (ii) turbulent broadening resulted in large systematic errors and large covariance matrix uncertainties. As a consequence, it was decided that AI-VPFIT should use compound broadening.
- Applied AI-VPFIT to the first ever LFC-calibrated quasar spectrum (produced by me) and obtained 40 new measurements of the fine structure constant (α) at $z = 1.15$ towards quasar HE0515–4414. The measurements present the first ever constraint on fundamental constant variations on kpc scales.
- Co-developed a new information criterion (SpIC) specifically designed for spectroscopy and to guide the AI-VPFIT modelling process, and tested its performance. Using SpIC results in 99% accuracy when retrieving simulated parameter values and results in no bias. For comparison, accuracy is 70% when other information criteria (e.g. AICc) are used.
- Published several other α measurements at high redshift and on in the presence of strong gravity (white dwarf surfaces).

ESPRESSO Large Programme for Fundamental Physics (7 peer-reviewed publications)

- Measured the relative carbon isotopic abundance at $z = 1.776$ towards quasar B1331+170 to be $^{12}\text{C}/^{13}\text{C} = 28.5^{+51.5}_{-10.4}$. For the first time ever, the quoted errors include uncertainties arising from model non-uniqueness. This robust result suggests an additional production of ^{13}C at low metallicity, in tension with many chemical evolution models.
- Investigated the origin of unexpected flux artefacts in ESPRESSO observations of faint objects. My work helped to identify them as weak warm pixels, allowing for the development of mitigation methods and the improvement of final data quality.
- Collaborated on improving ESPRESSO data reduction procedures.
- Published in the field of fundamental constants, inter-galactic medium, and isotopic abundances in metal-poor gas and stars.

ESO ELT Working Group on line calibrations

- Co-author of the ESO technical note “Recommendations and considerations stability of wavelength calibration for HIRES” (ESO-388314). A large part of recommendations were drawn from conclusions of my PhD research.
- Co-I of the approved ESO calibration proposal “An ESPRESSO iodine cell to prepare for ELT-HIRES science”. I contributed to all aspects, including writing the science case, defining instrument requirements and defining the observation strategy of the commissioning run.
- At the joint meeting of ESO ELT working groups in 03/2020, I presented the status of the field and discussed paths to overcome challenges required to reach ELT/ANDES goals.
- Invited to join the working group while still a PhD student (by M. Cirasuolo).

Scientific and strategic vision (2 peer-reviewed publications)

- Invited to write the section on redshift drift (IX.B.2) of the Snowmass 2021 contribution paper “Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies”.
- Co-author of the ESA Voyage 2050 contribution paper “Unveiling the faint ultraviolet Universe”.
- Member of Cosmoverse, a working group addressing observational tensions in cosmology with systematics and fundamental physics.
- Member of the Wide-field Spectroscopic Telescope (WST) working group on Cosmology.

Additional technical experience

- In-person participation in the commissioning of HARPS’s LFC in 09/2016.
- Co-authored reports of ESPRESSO Science Team to ESO on ESPRESSO / LFC performance.
- Invited to assess ESPRESSO LFC performance after a major intervention in 10/2022 (by A. Manescau).
- Developed a method to correct improper flux calibration of quasar spectra in the Sloan Digital Sky Survey Data Release 13.
- Developed software to simulate realistic high-resolution quasar spectra with provisions for input of instrument characteristics. It will be used to further refine methods and observing strategies for high precision measurements.

MANAGEMENT & RESPONSIBILITY ROLES

Leadership	Co-lead of a Working Package within WST Cosmology Working Group ESO student representative	since 2025 2017 – 2019
Reviewing	Invited to review for <i>Astrophysical Journal</i> , <i>Astrophysical Journal Letters</i> , <i>Scientific Reports</i> , <i>Advances in Space Research</i> , <i>Galaxies</i>	
Hiring committee	“Phi in the Sky” PhD grant allocation (Porto University). “Phi in the Sky” MSc grant allocation (Porto University).	2023 and 2024 2024
Administrative support	Organising Committee for the Munich Joint Astronomical Colloquium Assistant to the ESO Observing Programmes Committee	2019 – 2020 2017 and 2018
Teaching	“Astrocamp” lecturer, Vascões, PT. I developed and delivered 15h of lectures. Physics tutor for students aged 11-19 (150h)	2021 2015 – 2016
Supervision	Fabiana Pariti (undergraduate thesis) together with G. Cupani Julien Poyatos (undergraduate thesis) together with C. Martins.	2025 (expected) 2022
Organisation	Meetings of the ESPRESSO Working Group for Fundamental Physics IFPU workshop on fundamental constants, 10 participants, Trieste, IT IFPU Journal Club, Trieste, IT ESO Friday social event, Garching, DE ESO Science Day 2018, 100+ participants, Garching, DE ESO Journal Club, Garching, DE LOC chair for 3 rd IMPRS Student Symposium, Garching, DE LOC member “Multiwavelength-surveys: Galaxy Formation and Evolution from the early Universe to today”, Dubrovnik, HR	since 2020 2023 2022 – 2023 2017 – 2019 2018 2016 – 2018 2018 2015
Liaison & negotiation	Main liaison & coordinator between six partners in a private enterprise	since 2022

OTHER ACHIEVEMENTS & FUNDING

INAF technology grant (as Co-I, PI G. Cupani, project start Oct 2024)	100k EUR
ESO Early-Career Scientific Visitor Programme Garching, 6 weeks	3.5k EUR
ESO Director General’s Discretionary Fund, 4 weeks visit to IoA, Cambridge University, UK	2k EUR
International Max Planck Research School on Astrophysics, supported by ESO (Sep 2016 - Mar 2020)	80k EUR
CoSMass travel funds (awarded by project PI, V. Smolčić)	2k EUR
OzSTAR Supercomputer (Swinburne University of Technology, Melbourne, AU)	1.1M CPU-core hours
VLT/ESPRESSO (as Co-I on GTO and normal observing programmes)	>120h
ESO 3.6m/HARPS (as PI on a normal proposal)	8 nights
Featured as a talented young researcher in the ESO Annual Report 2017	

COLLABORATIONS & MEMBERSHIPS

Collaboration	EQUALS – Espresso QUasar Absorption Lines Survey (PI T. Berg)	since 2024
	ESPRESSO Large Programme on Fundamental Physics (PI P. Molaro)	since 2020
	B. Bassett (Cape Town), G. Cupani (Trieste), J. K. Webb (Cambridge)	
Membership	WST Cosmology working group (chairs M. Moresco and J.-P. Kneib)	since 2024
	Cosmoverse (chairs J. Levi Said and E. Di Valentino)	since 2023
	ESPRESSO Science Team (chair A. Sozzetti)	since 2020
	ESO ELT working group on Line Calibrations (coordinator C. Martins)	since 2019

TALKS & SEMINARS

Invited seminar	Swinburne University, Melbourne, AU	Dec 2024
Contributed talk	1 st meeting of INAF Computing Unit, Battaglia Terme, IT	Oct 2024
Contributed talk	17 th Marcel Grossmann meeting, Pescara, IT	Jul 2024
Contributed talk	Universum V, Rome, IT	Mar 2024
Invited talk	Spectral fidelity, Florence, IT	Sep 2023
Contributed talk	Cosmology at Miramare 2023, Trieste, IT	Aug 2023
Invited seminar	Weizmann Institute for Science, Tel Aviv, IL	May 2023
Invited review talk	Rencontres de Moriond, La Thuile, IT	Mar 2023
Seminar	ESO Lunch Talk, Garching, DE	Feb 2023
Contributed talk	ESPRESSO Science Team meeting, Lanzarote, ES	Feb 2023
Seminar	ESO Informal Discussion, Garching, DE	Jan 2023
Contributed talk	Tensions in Cosmology, Corfu, GR	Sep 2022
Invited talk	Hack 100, Trieste, IT	Jun 2022
Invited talk	The stubborn inconstancy of constants, Cambridge, UK	Nov 2021
Contributed talk	4 th Azores School of Cosmology, Angra do Heroismo, PT	Aug 2021
Invited seminar	OPINAS group, MPE, Garching, DE (online)	Apr 2020
Seminar	ESO Informal Discussion, Garching, DE	Mar 2020
Invited seminar	Trieste Observatory Colloquium, Trieste, IT	Jan 2020
Invited seminar	ESO Thirty minute talk, Santiago de Chile, CL	Dec 2018
Contributed talk	Shedding Light on the Dark Universe with ELTs, Trieste, IT	Jul 2018
Seminar	Department of Physics, Zagreb University, HR	Mar 2018
Contributed talk	4 th IMPRS Student Symposium, Garching, DE	Oct 2017
Contributed talk	3 rd Azores School of Cosmology, Angra do Heroismo, PT	Aug 2017

PUBLIC OUTREACH & ENGAGEMENT

Public talks	<i>Skeptics in the Pub</i> , Zagreb, HR (video has >500 views)	2024
	Petar Preradović Gymnasium, Virovitica, HR	2022
	<i>Astronomy on Tap</i> , London, UK (online, video has >100 views)	2021
	<i>Višnjan Summer School</i> , Višnjan, HR	2019
	<i>Skeptics in the Pub</i> , Zagreb, HR (video has >3600 views)	2018
Discussion panels	<i>Cinema and Science</i> , discussion on Artificial Intelligence, Trieste, IT	2024
Volunteering	Ambassador of Astronomy, IAU 100 funded project (proposer C. Harrison, link)	2019
	Ask an Astronomer, ESO Open Door day, Garching, DE	2017
	Open Door of the Ruđer Bošković Institute, Zagreb, HR	2015
Organisation	<i>Skeptics in the Pub</i> , >40 events with 50+ participants, Zagreb, HR	2011-2016
Interview	Vlog “Someone just like me” (video has >150 views)	2024
	Internet news portal ICV.hr (link)	2022
	ESO blog (link)	2020
	On <i>Skeptics in the Pub</i> for jutarnji.hr (link)	2014
Commentary	On the detection of GW170817 for index.hr (link).	2017
	The article was read by > 100k people (private communication with the editors).	

OTHER SKILLS

Language	Croatian	Native
	English	Proficient, C2
	Italian	Proficient, C1
	German	Basic, A2
Programming	Python (numpy, scipy, jax, jaxopt, numpyro)	Proficient
	Fortran	Intermediate
	git	Basic
Training	Public speaking & presentation	
	Courtesy, respect, and fair treatment	
	Mentoring & supervision	

Refereed journal articles

- 1 **D. Milaković**, J. K. Webb, P. Molaro, C.-C. Lee, P. Jethwa, G. Cupani, M. T. Murphy, L. Welsh, V. D’Odorico, S. Cristiani, R. Génova Santos, C. J. A. P. Martins, N. J. Nunes, T. M. Schmidt, F. A. Pepe, M. R. Zapatero Osorio, Y. Alibert, J. I. González Hernández, P. Di Marcantonio, E. Palle, et al.,
“Isotopic abundance of carbon in the DLA towards QSO B1331+170”, *MNRAS* **534**, 12–29 (2024).
- 2 F. Guarneri, L. Pasquini, V. D’Odorico, S. Cristiani, G. Cupani, P. Di Marcantonio, J. I. González Hernández, C. J. A. P. Martins, A. Suárez Mascareño, **D. Milaković**, P. Molaro, M. T. Murphy, N. J. Nunes, E. Palle, F. Pepe, R. Rebolo, N. C. Santos, R. G. Santos, T. M. Schmidt, S. G. Sousa, et al.,
“Fundamental physics with ESPRESSO: a new determination of the D/H ratio towards PKS1937-101”, *MNRAS* **529**, 839–854 (2024).
- 3 **D. Milaković** and P. Jethwa,
“A new method for instrumental profile reconstruction of high-resolution spectrographs”, *A&A* **684**, A38 (2024).
- 4 S. Cristiani, G. Cupani, A. Trost, V. D’Odorico, F. Guarneri, G. Lo Curto, M. Meneghetti, P. Di Marcantonio, J. P. Faria, J. I. González Hernández, C. Lovis, C. J. A. P. Martins, **D. Milaković**, P. Molaro, M. T. Murphy, N. J. Nunes, F. Pepe, R. Rebolo, N. C. Santos, T. M. Schmidt, et al.,
“Probing the small-scale structure of the intergalactic medium with ESPRESSO: spectroscopy of the lensed QSO UM673”, *MNRAS* **528**, 6845–6860 (2024).
- 5 P. Molaro, D. S. Aguado, E. Caffau, C. Allende Prieto, P. Bonifacio, J. I. González Hernández, R. Rebolo, M. R. Zapatero Osorio, S. Cristiani, F. Pepe, N. C. Santos, Y. Alibert, G. Cupani, P. Di Marcantonio, V. D’Odorico, C. Lovis, C. J. A. P. Martins, **D. Milaković**, M. T. Murphy, N. J. Nunes, et al.,
“The $^{12}\text{C}/^{13}\text{C}$ isotopic ratio at the dawn of chemical evolution”, *A&A* **679**, A72 (2023).
- 6 **D. Milaković**, C. C. Lee, P. Molaro, and J. K. Webb,
“Methods for quasar absorption system measurements of the fine structure constant in the 2020s and beyond”, *Mem. Soc. Astron. Italiana* **94**, 270 (2023).
- 7 C.-C. Lee, J. K. Webb, R. F. Carswell, V. A. Dzuba, V. V. Flambaum, and **D. Milaković**,
“Varying alpha, blinding, and bias in existing measurements”, *MNRAS* **521**, 850–865 (2023).
- 8 D. S. Aguado, E. Caffau, P. Molaro, C. Allende Prieto, P. Bonifacio, J. I. González Hernández, R. Rebolo, S. Salvadori, M. R. Zapatero Osorio, S. Cristiani, F. Pepe, N. C. Santos, G. Cupani, P. Di Marcantonio, V. D’Odorico, C. Lovis, N. J. Nunes, C. J. A. P. Martins, **D. Milaković**, J. Rodrigues, et al.,
“The pristine nature of SMSS 1605–1443 revealed by ESPRESSO”, *A&A* **669**, L4 (2023).
- 9 D. S. Aguado, P. Molaro, E. Caffau, J. I. González Hernández, M. R. Zapatero Osorio, P. Bonifacio, C. Allende Prieto, R. Rebolo, M. Damasso, A. Suárez Mascareño, S. B. Howell, E. Furlan, S. Cristiani, G. Cupani, P. Di Marcantonio, V. D’Odorico, C. Lovis, C. J. A. P. Martins, **D. Milaković**, M. T. Murphy, et al.,
“ESPRESSO observations of HE 0107–5240 and other CEMP-no stars with $[\text{Fe}/\text{H}] \leq -4.5$ ”, *A&A* **668**, A86 (2022).
- 10 R. Allart, C. Lovis, J. Faria, X. Dumusque, D. Sosnowska, P. Figueira, A. M. Silva, A. Mehner, F. Pepe, S. Cristiani, R. Rebolo, N. C. Santos, V. Adibekyan, G. Cupani, P. Di Marcantonio, V. D’Odorico, J. I. González Hernández, C. J. A. P. Martins, **D. Milaković**, N. J. Nunes, et al.,
“Automatic model-based telluric correction for the ESPRESSO data reduction software. Model description and application to radial velocity computation”, *A&A* **666**, A196 (2022).
- 11 V. da Fonseca, T. Barreiro, N. J. Nunes, S. Cristiani, G. Cupani, V. D’Odorico, A. C. O. Leite, C. M. J. Marques, C. J. A. P. Martins, **D. Milaković**, P. Molaro, M. T. Murphy, T. M. Schmidt, M. Abreu, V. Adibekyan, A. Cabral, P. Di Marcantonio, J. I. González Hernández, E. Palle, F. A. Pepe, et al.,
“Fundamental physics with ESPRESSO: Constraining a simple parametrisation for varying α ”, *A&A* **666**, A57 (2022).
- 12 E. Abdalla, G. F. Abellán, A. Aboubrahim, A. Agnello, Ö. Akarsu, Y. Akrami, G. Alestas, D. Aloni, L. Amendola, L. A. Anchordoqui, R. I. Anderson, N. Arendse, M. Asgari, M. Ballardini, V. Barger, S. Basilakos, R. C. Batista, E. S. Battistelli, R. Battye, M. Benetti, et al.,
“Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies”, *Journal of High Energy Astrophysics* **34**, 49–211 (2022).
- 13 C. J. A. P. Martins, S. Cristiani, G. Cupani, V. D’Odorico, R. Génova Santos, A. C. O. Leite, C. M. J. Marques, **D. Milaković**, P. Molaro, M. T. Murphy, N. J. Nunes, T. M. Schmidt, V. Adibekyan, Y. Alibert, P. Di Marcantonio, J. I. González Hernández, D. Mégevand, E. Palle, F. A. Pepe, N. C. Santos, et al.,
“Fundamental physics with ESPRESSO: Constraints on Bekenstein and dark energy models from astrophysical and local probes”, *Phys. Rev. D* **105**, 123507 (2022).

- 14 J. K. Webb, C.-C. Lee, and **D. Milaković**,
“Avoiding Bias in Measurements of Fundamental Constants from High Resolution Quasar Spectra”, *Universe* **8**, 266 (2022).
- 15 M. T. Murphy, P. Molaro, A. C. O. Leite, G. Cupani, S. Cristiani, V. D’Odorico, R. Génova Santos, C. J. A. P. Martins, **D. Milaković**, N. J. Nunes, T. M. Schmidt, F. A. Pepe, R. Rebolo, N. C. Santos, S. G. Sousa, M.-R. Zapatero Osorio, M. Amate, V. Adibekyan, Y. Alibert, C. Allende Prieto, et al.,
“Fundamental physics with ESPRESSO: Precise limit on variations in the fine-structure constant towards the bright quasar HE 0515–4414”, *A&A* **658**, A123 (2022).
- 16 **D. Milaković**, J. K. Webb, C.-C. Lee, and E. O. Zavarygin,
“Spectral shape corrections for SDSS BOSS quasars”, *A&A* **655**, A53 (2021).
- 17 C.-C. Lee, J. K. Webb, **D. Milaković**, and R. F. Carswell,
“Non-uniqueness in quasar absorption models and implications for measurements of the fine structure constant”, *MNRAS* **507**, 27–42 (2021).
- 18 C.-C. Lee, J. K. Webb, R. F. Carswell, and **D. Milaković**,
“Artificial intelligence and quasar absorption system modelling; application to fundamental constants at high redshift”, *MNRAS* **504**, 1787–1800 (2021).
- 19 A. Zanella, C. Zanoni, F. Arrigoni-Battaia, A. Rubin, A. F. Pala, C. Peroux, R. Augustin, C. Circosta, E. Emsellem, E. George, **D. Milaković**, R. van der Burg, and T. Kupfer,
“Unveiling the faint ultraviolet Universe”, *Experimental Astronomy* **51**, 913–943 (2021).
- 20 J. K. Webb, C.-C. Lee, R. F. Carswell, and **D. Milaković**,
“Getting the model right: an information criterion for spectroscopy”, *MNRAS* **501**, 2268–2278 (2021).
- 21 **D. Milaković**, C.-C. Lee, R. F. Carswell, J. K. Webb, P. Molaro, and L. Pasquini,
“A new era of fine structure constant measurements at high redshift”, *MNRAS* **500**, 1–21 (2021).
- 22 **D. Milaković**, L. Pasquini, J. K. Webb, and G. Lo Curto,
“Precision and consistency of astrocombs”, *MNRAS* **493**, 3997–4011 (2020).
- 23 M. R. Wilczynska, J. K. Webb, M. Bainbridge, J. D. Barrow, S. E. I. Bosman, R. F. Carswell, M. P. Dąbrowski, V. Dumont, C.-C. Lee, A. C. Leite, K. Leszczyńska, J. Liske, K. Marosek, C. J. A. P. Martins, **D. Milaković**, P. Molaro, and L. Pasquini,
“Four direct measurements of the fine-structure constant 13 billion years ago”, *Science Advances* **6**, eaay9672 (2020).
- 24 R. A. Probst, **D. Milaković**, B. Toledo-Padrón, G. Lo Curto, G. Avila, A. Brucalassi, B. L. Canto Martins, I. de Castro Leão, M. Esposito, J. I. González Hernández, F. Grupp, T. W. Hänsch, H. Kellermann, F. Kerber, O. Mandel, A. Manescau, E. Pozna, R. Rebolo, J. R. de Medeiros, T. Steinmetz, et al.,
“A crucial test for astronomical spectrograph calibration with frequency combs”, *Nature Astronomy* **4**, 603–608 (2020).
- 25 A. Butler, M. Huynh, J. Delhaize, V. Smolčić, A. Kapińska, **D. Milaković**, M. Novak, N. Baran, A. O’Brien, L. Chiappetti, S. Desai, S. Fotopoulou, C. Horellou, C. Lidman, and M. Pierre,
“The XXL Survey. XVIII. ATCA 2.1 GHz radio source catalogue and source counts for the XXL-South field”, *A&A* **620**, A3 (2018).
- 26 N. Baran, V. Smolčić, **D. Milaković**, M. Novak, J. Delhaize, F. Gastaldello, M. E. Ramos-Ceja, F. Pacaud, S. Bourke, C. L. Carilli, S. Ettori, G. Hallinan, C. Horellou, E. Koulouridis, L. Chiappetti, O. Miettinen, O. Melnyk, K. Mooley, M. Pierre, E. Pompei, et al.,
“The XXL Survey. IX. Optical overdensity and radio continuum analysis of a supercluster at $z = 0.43$ ”, *A&A* **592**, A8 (2016).

Unrefereed journal articles

- 1 P. Padovani, M. Cirasuolo, R. van der Burg, F. Cantalloube, E. George, M. Kasper, K. Leschinski, C. Martins, J. Milli, S. Möhler, M. Neeser, B. Neichel, A. Otavola, R. Sánchez-Janssen, B. Serra, A. Smette, E. Valenti, C. Verinaud, J. Vernet, O. Absil, et al., *The ESO’s Extremely Large Telescope Working Groups*, Dec. 2022.
- 2 M. T. Murphy, P. Molaro, T. M. Schmidt, C. J. A. P. Martins, V. da Fonseca, **D. Milaković**, G. Cupani, S. Cristiani, V. D’Odorico, T. Barreiro, R. Génova Santos, A. C. O. Leite, C. M. J. Marques, N. J. Nunes, F. A. Pepe, R. Rebolo, N. C. Santos, S. G. Sousa, M. R. Zapatero Osorio, M. Amate, et al., *ESPRESSO Probes the Fine-structure Constant*, Sept. 2022.

Under review & in print

- 1 C.-C. Lee, J. K. Webb, D. Dougan, V. A. Dzuba, V. V. Flambaum, and **D. Milaković**,
“Searching for new physics using high precision absorption spectroscopy; continuum placement uncertainties and the fine structure constant in strong gravity”, (2024), submitted to *Open Journal of Astrophysics*, 10.48550/arXiv.2410.01849.
- 2 J. K. Webb, C.-C. Lee, **D. Milaković**, D. Dougan, V. A. Dzuba, and V. V. Flambaum,
“Searching for new physics using high precision absorption spectroscopy; continuum placement uncertainties and $\Delta\alpha/\alpha$ towards the quasar PHL957”, (2024), submitted to *Open Journal of Astrophysics*, 10.48550/arXiv.2410.00684.

- 3 L. Pasquini and **D. Milaković**,
“Review of detector requirements: some challenges for the present”, (2024), a review article to appear in JATIS, 10.48550/arXiv.2405.14955.
- 4 J. K. Webb, C.-C. Lee, **D. Milaković**, V. V. Flambum, V. A. Dzuba, and J. Magueijo,
“The Mystery of Alpha and the Isotopes”, (2023), submitted to MNRAS Letters, 10.48550/arXiv.2401.00888.

Conference proceedings

- 1 A. Marconi, M. Abreu, V. Adibekyan, V. Alberti, S. Albrecht, J. Alcaniz, M. Aliverti, C. Allende Prieto, J. D. Alvarado-Gomez, C. S. Alves, P. J. Amado, M. Amate, M. I. Andersen, S. Antonucci, E. Artigau, C. Bailet, C. Baker, V. Baldini, A. Balestra, S. A. Barnes, et al.,
“ANDES, the high resolution spectrograph for the ELT: science goals, project overview, and future developments”,
Ground-based and airborne instrumentation for astronomy x, Vol. 13096, edited by J. J. Bryant, K. Motohara, and J. R. D. Vernet, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series (July 2024), p. 1309613.
- 2 **D. Milaković**,
“Fine structure constant measurements in quasar absorption systems”, Proceedings of the Moriond 2023 Gravitation conference (Oct. 2023), arXiv:2310.01071.
- 3 **D. Milaković**,
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